

Engineering Firms – How the Week Works

Read first

Your team is a real **engineering firm** hired to solve a real problem. You'll design a product, use AI to make the image and the slides, and **pitch it Friday to a review panel**. The grade isn't slide polish – it's solving a real problem and being able to **defend it**.

1 Your firm – 4 people, 4 roles, 4 deliverables

Lead Engineer

runs design decisions, keeps the firm on task

Turns in: Requirements sheet + the design decision (which idea, why, one trade-off)

Design Lead

owns the sketches and the AI product image

Turns in: 3 labeled sketches + the final design + the AI product image

Researcher

proves the problem is real

Turns in: Problem Evidence Brief – who has it, evidence, stakeholder

Presenter

runs Friday's pitch (everyone speaks, they lead)

Turns in: Pitch script (5 beats) + Q&A plan (who answers what)

Teams of 4 (a few of 3). Every person owns a deliverable – no passengers. Everyone speaks Friday.

2 Pick ONE problem (each firm takes a different one)

- | | |
|---|---|
| ☐ Backpacks wreck your shoulders and back. | ☐ Wet umbrellas soak everything indoors. |
| ☐ Phones die before the school day is over. | ☐ Charger cords are too short to reach your desk. |
| ☐ Water bottles leak inside a backpack. | ☐ Sports gear and cleats stink up your bag. |
| ☐ Nowhere on a desk for your phone, pencil, and earbuds. | ☐ Handouts get crushed or lost in a backpack. |
| ☐ Doors slam or won't stay open when your hands are full. | ☐ Water bottles won't stay cold all day. |
| ☐ Earbuds and charger cables tangle and disappear. | ☐ Bikes and scooters are a pain to lock up fast. |
| ☐ Lockers are a mess – stuff falls out every time. | ☐ Group work gets loud – no easy way to signal "we need help." |
| ☐ Hard to carry a lunch tray + a drink + your bag at once. | ☐ Your own (teacher-approved): _____ |

3 How you're graded

- **Your individual deliverable** (your role's artifact) – the real work is yours.
- **Team Design Review** – the 25-point rubric on the last page (problem, solution, how it works, defense, teamwork).

Monday – Start Your Firm

Mon • Day 6

Firm: _____ Your name + role: _____

TODAY YOU WILL

1. Get sorted into your firm and **assign the 4 roles**.
2. As a firm, **choose ONE problem** from the list (page 1).
3. Prove it matters: who has it, and how often?

1 Firm name

2 Members + roles

NAME	ROLE

3 Our problem + why it matters

Who has it, and how often?

Today's deliverable: firm formed, roles assigned, problem chosen, "why it matters" written. **Researcher** starts the Evidence Brief.

Tuesday – Define It Like an Engineer

Tue • Day 7

Firm: _____ Your name + role: _____

TODAY YOU WILL

1. Name exactly **who** has the problem and the **evidence** it's real.
2. Write **3–5 requirements** a good fix MUST meet – each one checkable (a number or yes/no).
3. Write your one-line **Success =** statement.

1 Who has it + evidence

2 Requirements (3–5, checkable – include one constraint: cost, size, or materials)

3 Success =

Not a requirement: "cool / nice / good." Rewrite it as something you can measure.

Today's deliverable: definition + requirements done. **Researcher's** Evidence Brief due; **Lead Engineer** owns the requirements.

Wednesday – Design It + Render It

Wed • Day 8

Firm: _____ Your name + role: _____

TODAY YOU WILL

1. **Each person sketches an idea** (at least 3 total). Make them different.
2. Score them against your requirements – pick one, write **why** + one **trade-off**.
3. Fill the prompt, generate the product image, fix it until it matches your design.

1 Three ideas

idea 1

idea 2

idea 3 – label the key part of each

2 Chosen design + why + one trade-off

Image prompt: "A clean product photo of a _____ that helps _____ by _____. Made of _____, about _____ big. Realistic, plain white background, studio lighting."

☐ AI product image generated and added to Google Classroom.

Today's deliverable: chosen design + AI image. **Design Lead** owns sketches + image; **Lead Engineer** records the decision.

Thursday – Build & Drill the Pitch

Thu · Day 9

Firm: _____ Your name + role: _____

TODAY YOU WILL

1. Use AI to build a **5-slide deck** from your packet (Problem → Evidence → Solution → How it works → Impact).
2. Assign the 5 beats – **everyone speaks**.
3. Drill the 4 tough questions – each person **owns one answer**. Do one full run.

1 Your 5-beat pitch

BEAT	YOUR ONE LINE	WHO SAYS IT
Problem		
Evidence		
Solution (+ image)		
How it works		
Impact		

2 Defense – who owns each answer?

THE PANEL WILL ASK...	WHO ANSWERS
Why this and not another idea?	
How does it actually work?	
What would it cost / what's it made of?	
Who exactly is this for?	

Today's deliverable: deck posted + practiced. **Presenter** owns the script + Q&A plan.

Friday – Design Review

Fri · Day 10

Firm: _____ Your name + role: _____

HOW TODAY RUNS

1. **3-minute pitch** – the 5 beats, everyone speaks.
2. **2-minute defense** – the panel asks, your firm answers.
3. Panel = Mr. Gavaldon + rotating student judges.

1 Rubric – 25 points

What we score	Points
Problem is clear and real	/5
Solution fits the requirements	/5
How it works makes sense	/5
Defense – answered the questions	/5
Teamwork – everyone contributed	/5
Total	/25

2 Peer judge (a firm that isn't yours)

Firm judged: _____ "What I liked is _____. My question is _____."

3 Reflection

One thing your firm did well, one thing you'd change.

Today's deliverable: presented + defended · all four individual deliverables turned in.